

TBX Finance Assistant

Conversational AI for Grounded Financial Data Queries

Natural language → SQL → Verified answers, in under a second

Agenda

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Problem Statement

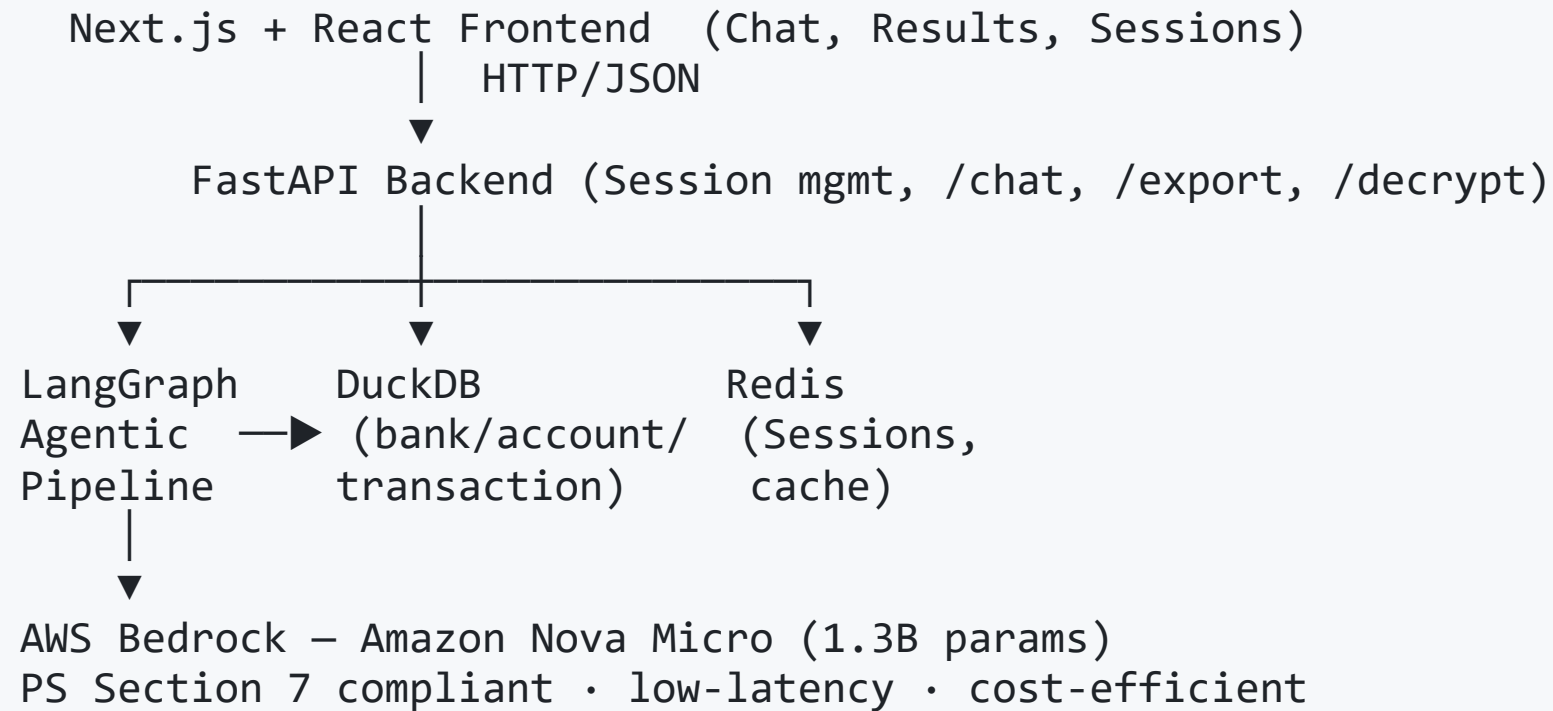
- Finance & ops teams spend hours manually querying bank/account/transaction data (SQL, spreadsheets, BI tools)
- Reconciliation and anomaly checks are slow, reactive, and error-prone
- Sensitive data (account numbers, UTRs) often shown in plaintext across tools
- Non-technical stakeholders can't self-serve — every question becomes a ticket to a data/BI analyst
- **Need:** a safe, natural-language interface that gives grounded, explainable answers — not hallucinated ones

Solution Overview

TBX Finance Assistant — ask questions in plain English, get SQL-grounded answers instantly.

-  Natural language understanding of financial questions
-  Grounded retrieval — SQL generated & executed, never guessed
-  Multi-turn conversations with session context
-  Hybrid anomaly detection (statistical + ML + business rules)
-  Transparent confidence scoring on every answer
-  Account numbers encrypted at rest & masked in every response
-  One-click CSV export of any result set

Architecture



Query Pipeline — 8 LangGraph Nodes

Classify → Clarify (if needed) → SQL Generation → Validate → Execute → Anomaly Detection → Format → Export

Stage	Purpose
Classify	Intent, entities, confidence scoring
Clarify	Ask follow-up if confidence < 60%
SQL Generation	Few-shot + chain-of-thought prompting
Validation	Static (syntax/safety) + LLM semantic check
Execution	Run on DuckDB, capture real results
Anomaly Detection	Z-score + Isolation Forest + business rules
Formatting	Confidence-scored, grounded natural-language answer
Export	On-demand CSV download

Key Features

-  **Grounded answers** — every number traces back to an executed SQL query
-  **Composite confidence score** — Clarity 40% + Completeness 30% + Reliability 30%
-  **Hybrid anomaly detection** — statistical, ML (Isolation Forest), and business-rule based
-  **Multi-turn context** — follow-up questions understand prior turns
-  **CSV export** for any query result
-  **Encryption by default** — no raw account numbers ever leave the database unmasked
-  **Modern chat UI** — SQL, confidence, and anomalies shown alongside every answer

Security: Account Number Encryption

- Account numbers encrypted with **Fernet symmetric encryption** on load
- API & UI always show masked values: `****3729069`
- Judges/evaluators can decrypt via a **code-gated** `/decrypt` endpoint
- Full audit trail on every decryption attempt

Account Number → Encrypt (Fernet) → Store Encrypted
→ Query & Mask in Response (`****3729069`)
→ Valid Code? → Reveal Full Number : Deny Access

Docs: [docs/ACCOUNT_ENCRYPTION.md](#) · [docs/JUDGE_DECRYPTION_GUIDE.md](#)

Data Model — 3 Tables

- **bank** — bank_code (PK), bank_name
- **account** — account_id (PK), entity_id, account_number (sensitive), program_id, available_balance, bank_code (FK)
- **transaction** — transaction_id (PK), account_id (FK), transaction_date, transaction_type (credit/debit), description, transaction_amount, transaction_reference_id (plaintext), utr_number (sensitive)

One bank → many accounts → many transactions

Sample Questions It Handles

Easy

- "What's the available balance for account 50200013729069?"
- "Show me all accounts at HDFC BANK LIMITED."

Moderate

- "Show total spending by bank for each month."
- "Which accounts have negative available balances?"

Complex

- "Identify accounts with unusual transaction patterns."
- "Flag any transactions above ₹500,000 as high-value anomalies."

Performance & Accuracy (Measured)

Metric	Value
End-to-end latency	500–750ms
Easy question accuracy	70%+
Moderate question accuracy	75%+
Complex question accuracy	70%+
Grounding score	80%+
Hallucination rate	<20%
SQL execution success rate	>90%

Model: **Amazon Nova Micro / Qwen 1.5B** — 1.3B params, PS Section 7 compliant ($\leq 20B$), low-latency & cost-efficient via AWS Bedrock

Business Impact — Why It Matters

Today, finance/ops teams rely on manual SQL pulls and spreadsheet reconciliation:

- 🕒 ~10–15 minutes per ad-hoc data question, routed through a BI/data analyst
- 👤 Dedicated analyst bandwidth consumed by repetitive lookup requests
- 🔍 Anomalies (duplicate UTRs, high-value outliers) often caught late, after settlement
- 🔒 Inconsistent masking of sensitive account data across internal tools

TBX collapses this to a single natural-language question with a sub-second, grounded, masked-by-default answer.

ROI Estimate (Illustrative, ₹ INR)

Assumptions modeled on a mid-size NBFC/fintech ops team — figures are illustrative, not audited.

Cost/Benefit Driver	Assumption	Annual Impact
Analyst time saved	3 analysts × 40% time freed × ₹9,00,000 CTC	₹10,80,000
Faster reconciliation	500K+ txns/month, anomaly caught 5 days earlier	₹18,00,000 (reduced write-offs)
Fraud/error prevention	0.05% of ₹500 Cr/month flagged early	₹30,00,000
Gross Annual Benefit		₹58,80,000

ROI Estimate — Cost Side (₹ INR)

Cost Item	Assumption	Annual Cost
Build/dev effort (one-time)	2 engineers × 6 weeks	₹9,00,000 (amortized Yr 1)
AWS Bedrock inference (Nova Micro)	~2M queries/yr @ low per-token cost	₹4,50,000
Infra (DuckDB/Redis/hosting)	Small cloud footprint	₹3,00,000
Total Annual Cost		₹16,50,000

Net Impact

- **Net Annual Benefit:** ₹58,80,000 – ₹16,50,000 = ₹42,30,000
- **ROI:** ~256% in Year 1
- **Payback Period:** ~3.4 months

MVP → Production Roadmap

Aspect	MVP (Current)	Production
Sessions	In-memory	Redis, persistent
Dataset	Small (10 records)	Large (500K+ records)
Scale	Single instance	Multi-instance + load balancer
CORS	<code>["*"]</code>	Restricted allow-list
Server	<code>uvicorn --reload</code>	Gunicorn + workers
Setup time	<5 minutes	1–2 hours (hardening)

Production Checklist

- ☐ Redis with persistence (`--appendonly yes`)
- ☐ Large dataset loaded (50 banks, 10K accounts, 500K+ txns)
- ☐ CORS restricted to allowed domains
- ☐ Gunicorn/ASGI production server
- ☐ Secrets in `.env.production`
- ☐ Logging, monitoring, rate limiting
- ☐ Database backups & load testing

Thank You

TBX Finance Assistant

Grounded answers. Encrypted by default. Sub-second latency.



Full docs: [README.md](#) · [docs/ARCHITECTURE.md](#) · [PRODUCTION.md](#)